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Day 08 Post-Session Assignment & Engineering Reflection

AI Engineering Program | Day 08

Part 1 - Evaluation evidence

- **Evaluation Dataset**: Established a reusable evaluation dataset (day08_cases.json) comprising 12 test cases spanning direct facts, paraphrases, conditional rules, missing details, unsupported policies, comparison queries, and boundary conditions.
- **Baseline Results (day08_baseline.json)**:
  Executed the 12-case evaluation suite using the RAG pipeline with Top-K = 3 and local llama3.
  Pass/Fail Summary: **9 passed out of 12** cases.
  Failures Observed: Cases D08-05 (Missing Detail), D08-07 (Supported + Unsupported Comparison), and D08-11 (Policy Exception) failed due to answer relevance and insufficient evidence handling constraints.

Part 2 - Failure analysis

- **Most Common Failure Category**: **Answer Relevance** and **Missing Evidence Handling** (tied for primary failure drivers where the model either retrieved related text but answered a slightly different question—such as confusing finance approval thresholds with receipt thresholds—or failed to explicitly flag partial information gaps).
- **Highest Business Impact**: **Missing Detail / Unsupported Policy Handling (D08-05 & D08-07)**. Failing to recognize when an exact monetary receipt threshold or a parental-leave duration is missing from the text risks hallucinating compliance rules or policy fabrications that could lead to financial or HR policy violations.
- **Failure That Looked Like a Prompt Problem**: Case D08-11 (*Does an open support ticket automatically extend temporary access?*). Initially suspected to be a prompt instruction failure regarding negative constraints, diagnosis revealed it was an **output validation / formatting constraint clash** where the model's structured JSON output schema struggled with conditional abstention fields.
- **Most Useful Test Case**: **Case 5 (D08-05: Missing Detail)**. Asking for an exact monetary receipt threshold when the text only mentions finance approval limits (INR 10,000) exposed a classic RAG trap: the model successfully retrieved a real source quote and cited it correctly, but used it to answer the *wrong* question (answer relevance failure).

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Part 3 - Controlled improvement

- **ONE Controlled Change Made**: Implemented an experimental response contract (day08_improved.json) allowing partial-evidence handling (missing_parts) via prompt adjustments, while keeping retrieval, chunks, model, and decoding temperature (0) strictly fixed.
- **Evaluation Evidence Justification**: Baseline evaluation showed that queries asking to compare supported vs. unsupported topics (like annual vs. parental leave) resulted in total abstention, discarding valid retrieved facts. The change aimed to preserve supported facts while explicitly flagging missing details.
- **Which Cases Improved**: None of the target cases successfully transitioned to passing.
- **Did Any Cases Regress**: **Yes**. Passing cases dropped from **9/12 to 6/12**. Cases D08-03, D08-06, and D08-10 regressed due to schema validation rejections caused by inconsistent abstention fields and partial-label constraints.
- **Would You Keep the Change?**: **No**. As documented in day08_comparison.md, we rejected the candidate change and **kept the baseline**. An experimental feature that reduces passing test coverage and introduces structural JSON validation errors is not production-ready.

Part 4 - Engineering reasoning

- **Why a successful demo is insufficient evidence of RAG quality**: A demo relies on cherry-picked "happy path" questions. Systematic evaluation using a diverse test harness reveals hidden failure modes—such as semantic drift, boundary confusion, and failure on unsupported queries—that casual testing misses.
- **Why retrieval and answer quality should be evaluated separately**: A system can successfully retrieve the exact correct text chunk (Rank 1) yet still fail if the LLM misinterprets the text, hallucinates an answer, or answers a different question. Separating retrieval metrics (hit/miss) from generation metrics (groundedness/relevance) is essential for root-cause debugging.
- **Why one aggregate score can hide important risk**: An aggregate score (e.g., 75% overall pass rate) obscures catastrophic failures. Passing 9 easy direct-fact questions while failing high-risk compliance or missing-detail checks creates a dangerous false sense of security.
- **What should trigger regression evaluation in a real system**: Any modification to document parsers, chunking strategies, embedding models, vector store indexing parameters, prompt system instructions, or base LLM versions must automatically trigger full evaluation suite regression testing.

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Part 5 - FDE reflection

- **Customer-Unacceptable Failure Types**: Customers in legal, HR, or finance domains will find **ungrounded hallucinations, false citations, and invented policy thresholds** (e.g., guessing receipt monetary limits) completely unacceptable.
- **Who Should Define Domain-Specific Expected Behavior**: Domain subject matter experts—specifically HR policy managers, finance leads, and security/compliance officers—must collaborate with engineers to establish the exact expected answers and abstention rules for edge cases.
- **Recommendation (Iteration vs. Pilot vs. Stop)**: **Recommend another targeted iteration**, not a production pilot yet. While core factual retrieval is strong, handling missing details and partial information requires refining response schemas and prompt rigidity before exposing the tool to end users.
- **Evidence Supporting Recommendation**: Present the comparative evaluation report (day08_comparison.md), showing that baseline RAG achieves 75% stability (9/12) on structured unit test cases, but detailing the exact regression risks and validation failures encountered during tuning.

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